

Modeling and Testing of Hydrodynamic Damping Model for a Complex-shaped Remotely-operated Vehicle for Control

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Abstract: In this paper, numerical modeling and model testing of a complex-shaped remotely-operated vehicle (ROV) were shown. The paper emphasized the systematic modeling of hydrodynamic damping using the computational fluid dynamic software ANSYS-CFXTM on the complex-shaped ROV, a practice that is not commonly applied. For initial design and prototype testing during the developmental stage, small-scale testing using a free-decaying experiment was used to verify the theoretical models obtained from ANSYS-CFXTM. Simulation results are shown to coincide with the experimental tests. The proposed method could determine the hydrodynamic damping coefficients of the ROV.

Keywords: remotely-operated vehicle; hydrodynamic damping; ANSYS-CFXTM; modeling; simulation

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1 Introduction

Modeling and simulation lies in the heart of most current technological innovations and has become a fundamental tool in many fields of engineering. Essentially multi-disciplinary in its applications, modeling and simulation is in its own right a key technology which is bound to increase its presence within industries and even institutions. The proposed work on a marine vehicle such as an underwater robotic vehicle (URV) reflects this multidisciplinary, drawing on the unifying mathematical aspects from separate engineering disciplines.

The applications of this multidisciplinary URV (Jalving, 1994; Healey and Lienard, 1993; Gomes *et al.*, 2003; Antonelli *et al.*, 2001; Fjellstad and Fossen, 1994; Fossen, 1994; Fossen, 2002) design have experienced tremendous growth in the last few decades. Many are used for underwater inspection of sub-sea applications such as installation of structures and pipelines. They are essential at depths where the use of human divers is impractical. Based on the designed tasks and modes of operations, the URVs are broadly classified as remotely operated vehicles (ROV) and autonomous underwater vehicles (AUV). Nevertheless, the ROVs are best suited for work that involves operating from a stationary point or cruising at relatively slow speeds such as for pipeline inspection where the hard wire link is still required. For any tasks involving manipulation and requiring maneuverability, they are the most cost-effective platform. Besides the application aspect of the ROV, it is difficult to model (Goheen, 1991; An and Folleco, 2003; Gomes *et al.*, 2003) and control the complex-shaped ROV as the body design is often asymmetrical and sensors are attached

to its inherent open-frame structure as compared to its counterpart, the AUV, which is torpedo-shaped for faster maneuvering. Additionally, the modeling task is alleviated by the nonlinearity and uncertain vehicle dynamics.

The vehicle dynamics are often obtained using different kinds of test equipment. Most of the dynamic testing is conducted on the model undergoing forced lateral or vertical plane motions to determine damping as well as some other derivatives such as added mass; for surface models these are invariably frequency-dependant. Routine dynamic testing was introduced with the planar motion mechanism (PMM) (Goodman, 1960). The PMM imparts harmonic oscillations to the model. Vertical PMMs for subsurface vehicles are, with few exceptions, low amplitude devices for obtaining linear hydrodynamic coefficients; horizontal PMMs more often have large amplitude. A new generation of the test apparatus has been represented by the Marine Dynamic Test Facility (Williams *et al.*, 2000). This device imparts large amplitude and high rate arbitrary motions to a subsurface or surface model in up to six degrees-of-freedom (DOF), and the device contains a 6-DOF force sensor to measure total forces and moments in water. On the other hand, predictive methods offer an attractive alternative to test-based methods when the vehicle is still in the design stages or when cost prohibits a full-scale testing program. The most basic of the predictive methods are purely analytical methods, but these are prone to yielding unrealistic results for complex bluff bodies. Jones *et al.* (2002) provided a detailed discussion and evaluation of the existing methods. Although the hydrodynamic damping for the ROV is usually tested by the above-mentioned full-scale tow tank facilities, it could be difficult to justify building the facility just for the ROV testing. Besides, for initial design and prototype testing, smaller-scale testing is often desirable and economical to run during the developmental stage. The immediate

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challenge to design a control system for a marine vehicle is to obtain an appropriate mathematical model. With the increasing development of computers, applications of computational-fluid dynamics (CFD) in the maritime industry have become important. Numerical approaches have evolved to a level of accuracy, which allows them to be applied to marine vehicles such as the streamline-shaped AUV (Sarkar *et al.*, 1997; Tyagi and Sen, 2006; Wilson *et al.*, 2006). However, application of the complex-shaped ROV is not yet available in research literature.

In this paper, a semi-predictive approach of using computer-aided design (CAD) software PRO/ENGINEER™ to draw the ROV while using the CFD software ANSYS-CFX™ to obtain the hydrodynamic damping forces is presented. The results obtained by this method are compared with experimental study on a scale model of the ROV in a small water tank that is later translated to the full-scale model by Laws of Similitude (Conte *et al.*, 2004). The free-decaying test as shown here can be used for medium and small class underwater vehicles and could be a viable alternative to estimate some pertinent hydrodynamic parameters without extensive and expensive facilities and instrumentation at a very early stage of the ROV development.

In summary, the proposed method provides a convenient interface with commonly used hydrodynamic codes that are widely adopted during preliminary stages of the ship design and later on the complex-shaped ROVs. The proposed method provides most of the parameters necessary to implement a model of the ROV that could be used as an initial step in the control system design process. With the proposed experimental method using the free-decaying test, the systematic approach to model the hydrodynamic damping coefficients of the ROV will be a viable option in the near future.

This paper is organized as follows. The ROV dynamics are addressed in Section 2. In Section 3 the hydrodynamic damping modeling process and comparisons with the theoretical results are presented. In Section 4, the experimental results of the damping coefficient using the free-decaying method is described. Lastly, the conclusions are drawn in Section 5.

2 Dynamics of ROV

Modeling dynamic equations of the ROV is usually the first step in developing a simulation. In this section, the basic design of the ROV is described. The ROV designed by the Robotics Research Centre (RRC) in Nanyang Technological University (NTU) is known as the RRC ROV (see Fig.1). It is used to perform underwater pipeline inspections such as locating pipe leakages or cracks. Since pipelines carry hydrocarbons such as natural gas or oil at relatively high pressures, a crack in the line can result in an explosion as

well as the accidental discharge of oil and/or gas with adverse environmental consequences. It is therefore, desirable to inspect the pipelines periodically and to perform the tasks efficiently. This can be done with the operator focusing on inspection while the ROV automatically tracks the pipeline.

The twin “eye-ball” ROV depicted in Fig.1 has an open-frame structure and is 1 m long, 0.9 m wide, and 0.9 m high. It has a dry weight of 115 kg and a current operating depth of 100 m. Its designed tasks include inspections of underwater pipelines. The RRC ROV is underactuated as it has four thruster inputs for six degree of freedom (DOFs) (that is surge, sway, heave, roll, pitch, and yaw velocity) with a high degree of cross coupling between them. The vehicle is equipped with two lateral thrusters, mainly for surge, sway, and yaw motion control in the horizontal plane, two vertical thrusters for the heave in the vertical plane, and a suite of sensors for position and velocity measurements. Roll and pitch motions are passive as the metacentric height (that is the distance between the center of gravity of a ROV and its metacenter) is sufficiently large to provide adequate static stability. A brief description of the component layout of the RRC ROV is given.

- 1) Four thrusters, each providing up to 70N of thrust;
- 2) Two cylindrical floats with four balancing steel weight;
- 3) Main pod (Pod 1) and sensors with navigational pod (Pod 2) and external sensor such as altimeter.

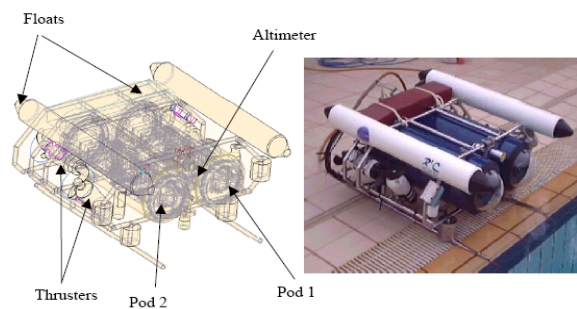


Fig.1 Schematic of RRC ROV

In general, there are basically two methods to derive the dynamic model: the Lagrange equations using the variation method and Newtonian equations using Newton's Second Law of Motion. Euler's transformation (commonly known as the kinematics equation) provides the relationship between the dynamics derived in the earth-fixed, and the body-fixed coordinates. There are practical advantages (An and Folleco, 2003) to derive the dynamics in the body-fixed coordinate. First, most sensors and actuators mounted on the ROV measure the body parameters such as the ROV velocities and provide propelling forces. Secondly, the dynamic equation is inherently parameterized and hence less complicated since this parameterized form is not dependent on Euler's angles. Prior to ROV modeling, the following assumptions can be made when deriving the general ROV dynamic equation in

order to simplify the effort in modeling. They are namely:

- (a) The ROV is a rigid body and is fully submerged once in water;
- (b) Water is assumed to be an ideal fluid that is incompressible, inviscid (frictionless), and irrotational;
- (c) The ROV is slow moving for operations such as pipeline inspection;
- (d) The earth-fixed frame of reference is inertial;
- (e) Disturbance due to waves is neglected as it is fully submerged;
- (f) Tether dynamics attached to the ROV are not modeled.

The standard Society of Naval Architects and Marine Engineers (SNAME) notations used for the marine vehicle are shown in Table 1.

Table 1 Notations used in ROV

DOF	Motion descriptions	Positions and orientations	Linear and angular velocities
1	Motions in the x-direction(surge)	x	u
2	Motions in the y-direction(sway)	y	v
3	Motions in the z-direction(heave)	z	w
4	Rotations about the x-axis(roll)	ϕ	p
5	Rotations about the y-axis(pitch)	θ	q
6	Rotations about the z-axis(yaw)	ψ	r

2.1 ROV dynamic equations

A rigid body ROV dynamic equation is expressed in the body-fixed frame as the control forces and measurement devices are intuitively related to this body frame of reference. Using the Newtonian approach, the motion of a rigid body with respect to the body-fixed reference frame at the origin (see Fig.2) is given by the equation (An *et al.*, 2003):

$$\mathbf{M}_{RB}\dot{\mathbf{v}} + \mathbf{C}_{RB}(\mathbf{v}) = \boldsymbol{\tau} \quad (1)$$

where, $\mathbf{M}_{RB} \in \mathbb{R}^{6 \times 6}$ is the mass-inertia matrix, $\mathbf{C}_{RB}(\mathbf{v}) \in \mathbb{R}^{6 \times 6}$ is the Coriolis and centripetal matrix, $\boldsymbol{\tau} = [\boldsymbol{\tau}_1 \ \boldsymbol{\tau}_2]^T \in \mathbb{R}^6$ is a vector of external forces and moments, $\mathbf{v} = [\mathbf{v}_1 \ \mathbf{v}_2]^T \in \mathbb{R}^6$ is the linear and angular velocity vector, namely: $\mathbf{v}_1 = [u \ v \ w]^T \in \mathbb{R}^3$ and $\mathbf{v}_2 = [p \ q \ r]^T \in \mathbb{R}^3$ respectively.

The mass inertia matrix given in Eq.(1) can be written as:

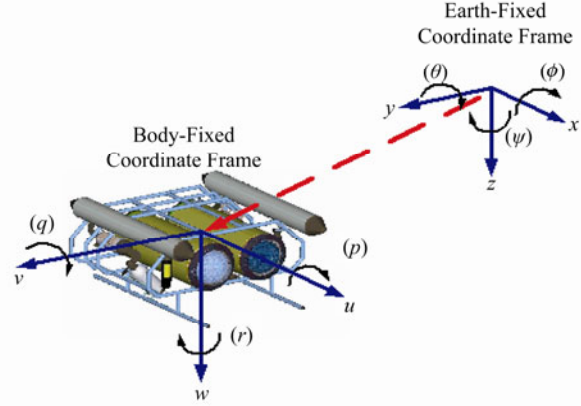


Fig.2 Coordinate systems used in ROV

$$\mathbf{M}_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_G & -my_G \\ 0 & m & 0 & -mz_G & 0 & mx_G \\ 0 & 0 & m & my_G & -mx_G & 0 \\ 0 & -mz_G & my_G & I_x & -I_{xy} & -I_{xz} \\ mz_G & 0 & -mx_G & -I_{yx} & I_y & -I_{yz} \\ -my_G & mx_G & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix} \quad (2)$$

Similarly, the Coriolis and centripetal terms, describing the angular motion of the ROV can be expressed as:

$$\mathbf{C}_{RB}(\mathbf{v}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{C}_{12}(\mathbf{v}) \\ -\mathbf{C}_{12}^T(\mathbf{v}) & \mathbf{C}_{22}(\mathbf{v}) \end{bmatrix} \quad (3)$$

with

$$\mathbf{C}_{12}(\mathbf{v}) = \begin{bmatrix} m(y_G q + z_G r) & -m(x_G q - w) & -m(x_G r + v) \\ -m(y_G p + w) & m(z_G r + x_G p) & -m(y_G r - u) \\ -m(z_G p - v) & -m(z_G q + u) & m(x_G p + y_G q) \end{bmatrix} \quad (4)$$

$$\mathbf{C}_{22}(\mathbf{v}) = \begin{bmatrix} 0 & -I_{yz}q - I_{xz}p + I_z r & I_{yz}r + I_{xy}p - I_y q \\ I_{yz}q + I_{xz}p - I_z r & 0 & -I_{xz}r - I_{xy}q + I_x p \\ -I_{yz}r - I_{xy}p + I_y q & I_{xz}r + I_{xy}q - I_x p & 0 \end{bmatrix} \quad (5)$$

The external force and moment vector $\boldsymbol{\tau}$ includes the hydrodynamic forces and moments, $\boldsymbol{\tau}_H$ due to damping and inertial of surrounding fluid known as added mass, and the restoring force and moment. These forces and moments tend to oppose the motion of the ROV and, other than the restoring force, are dependent on the velocities and accelerations of the vehicle and are hence expressed in the body-fixed frame. The restoring force and moment due to the weight and buoyancy force have to be transformed to the body-fixed frame forces and moments $\mathbf{G}_f(\boldsymbol{\eta})$. The open-loop nonlinear ROV dynamic equations can be expressed according to:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_A + \boldsymbol{\tau}_H = \boldsymbol{\tau}_A - \mathbf{M}_A - \mathbf{C}_A(\mathbf{v}) - \mathbf{D}(\mathbf{v})\mathbf{v} - \mathbf{G}_f(\boldsymbol{\eta}) \quad (6)$$

The mass inertia matrix given in Eq.(1) can be written as:

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{G}_f(\boldsymbol{\eta}) = \boldsymbol{\tau}_A \quad (7)$$

where $\mathbf{v} = [u \ v \ w \ p \ q \ r]^T$ is the body-fixed velocity vector and $\boldsymbol{\eta} = [\boldsymbol{\eta}_1 \ \boldsymbol{\eta}_2]^T$ is the earth-fixed vector, comprising

the position vector $\eta_1 = [x \ y \ z]^T \in \mathbb{R}^3$ and the orientation vector of Euler angles, $\eta_2 = [\phi \ \theta \ \psi]^T \in \mathbb{S}^3$ (a torus of dimensions three); the angles are defined on the interval $[0, 2\pi)$. $M = M_{RB} + M_A \in \mathbb{R}^{6 \times 6}$ is the sum of the rigid body inertia mass and added fluid inertia mass matrix, $C(v) = C_{RB}(v) + C_A(v) \in \mathbb{R}^{6 \times 6}$ is the sum of Coriolis and centripetal and the added mass forces and moments matrix, and $D(v) \in \mathbb{R}^{6 \times 6}$ is the damping matrix due to the surrounding fluid. The input force and moment vector $\tau_A = Tu \in \mathbb{R}^6$ relates the thrust output vector $u = F_T \bar{u} \in \mathbb{R}^4$ with the thruster configuration matrix $T \in \mathbb{R}^{6 \times 4}$, $F_T \in \mathbb{R}^{4 \times 4}$ is the dynamics of each thruster and converts the input voltage command $\bar{u} \in \mathbb{R}^4$ into thrusts to propel the vehicle.

3. Hydrodynamic damping model

In this section, the procedures of finding the hydrodynamic damping coefficients of the ROV are explained. First, the CAD software PRO/ENGINEER™ (see Fig.3) was used to determine the rigid-body mass and inertia of the ROV. The principal components were included in the complete RRC ROV geometric model and using the density and mass values available, the rigid-body mass and inertia properties with respect to the ROV's center of gravity can be determined.

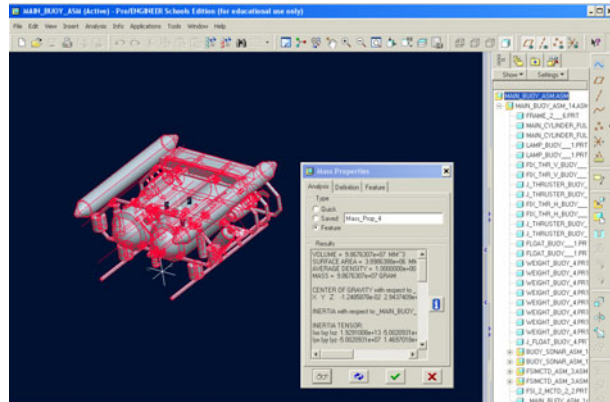


Fig. 3 CAD software PRO/ENGINEER™ for ROV

By adding balancing weights at designated location on the ROV, the location of the center of gravity can be made to coincide with the ROV origin. The parameters used in (2) become:

$$m=115.00 \text{ kg}, I_x=6.1000 \text{ kg}\cdot\text{m}^2, I_y=5.9800 \text{ kg}\cdot\text{m}^2, \\ I_z=5.5170 \text{ kg}\cdot\text{m}^2, I_{xy}=-0.0002 \text{ kg}\cdot\text{m}^2, I_{xz}=-0.1850 \text{ kg}\cdot\text{m}^2, \\ I_{yz}=0.0006 \text{ kg}\cdot\text{m}^2,$$

The matrix D is the hydrodynamic damping matrix consisting of both linear and quadratic terms. This hydrodynamic damping is caused by the potential damping due to linear skin friction (linear damping), drag, and vortex shedding (quadratic or nonlinear damping). The sum of these individual components gives the overall hydrodynamic

damping effect on the ROV.

Since the ROV is designed to be self-stabilizing in the roll and pitch angle, the magnitude of these angles especially the pitch angle is very small. Hence for such a small pitch angle (or angle of attack), there is no cross-flow separation and no boundary-layer separation, and the flow remains attached. Therefore, the nonlinear parts of the forces and moments can be considered as forces and moments due to viscous effect of the flow, which becomes less important as the pitch angle is smaller. In this situation, the body essentially behaves like a lifting surface, and the forces and moments can be determined from inviscid flow theory. These potential flow forces have a linear relationship with the pitch angle. Thus, it may be stated that the linear coefficients are sufficient to represent the force and moments arising from the inviscid part of the flow, in particular, for slow speed (less than 2m/s) (An *et al.*, 2003) maneuvering of the ROV and with minimal pitching during maneuvering. For comparison purposes, both linear and nonlinear damping forces were computed.

As observed, the ROV is a complex block structure. This may suggest that a more complete approach is required to quantify the hydrodynamic coefficients. An alternative semi-predictive approach that uses the open-tank and the CFD method using ANSYS-CFX™ is used. This method is becoming increasingly tractable due to the development of CFD packages and the advancement of computer technology such as the use of MATLAB™ to facilitate the determination of the parameters obtained from the tank test and then use of ANSYS-CFX™ to verify the hydrodynamic coefficients. However, in any methods used, it will always be subject to experimental and numerical errors (WS Atkins Consultants, 2002). Thus, due to this error or uncertainties, the model derived by either method has to depend on some form of a robust control scheme to control the ROV. The hydrodynamic damping coefficients of the scaled ROV (Eng, 2007; Eng *et al.*, 2009; Eng *et al.*, 2008) were obtained using ANSYS-CFX™ software. The accuracy of the numerical approach using ANSYS-CFX™ was established through comparison of the current study with empirical and other computational results of the sphere. These results could be truly determined if the results could be independently confirmed using a different approach. The damping coefficients were verified using the free-decaying experiment on the scaled ROV. By applying laws of Similitude, the hydrodynamics parameters of the scaled model can be scaled up to predict the corresponding values for the true model (or the actual ROV model).

Using the assumptions (An *et al.*, 2003), the hydrodynamic damping matrix can be simplified by using the following assumptions. (a) As the ROV is operating at around a linear speed of 0.5 m/s (or angular speed of 0.5 rad/s), it is well within the linear damping region maximum of 2 m/s (An *et*

al., 2003). Hence, the linear hydrodynamic damping was considered. But for completeness, the quadratic hydrodynamic damping force was determined. (b) The off-diagonal elements in $\mathbf{D}$ on a typical underwater vehicle are small compared to the diagonal elements; hence only the hydrodynamic damping in the diagonal form is used:

$$\mathbf{D} = -\text{diag}\{X_u, Y_v, Z_w, K_p, M_q, N_r\} \quad (8)$$

Like the added mass terms, the hydrodynamic damping force and moment coefficients for the ROV are difficult to obtain experimentally without a proper full scale instrumented tow tank facility. However, it could be expensive and difficult to justify building the facility for this usage only. The problem is circumvented using ANSYS-CFXTM and an experimental study of a scaled model in a small water tank that is later translated to the full-scale model by using laws of Similitude (Conte *et al.*, 2004).

In CFD modeling, the physical model for computer simulation has to be determined. Figure 4 shows the ROV moving forward at a constant speed while the fluid domain remains static with no current flow. Instead of the ROV moving, the flow can be made to move at a constant speed in the opposite direction as the ROV which remains static in position. Since the drag force depends only on the relative motion between the ROV and the fluid, the result obtained from the two situations is the same. The advantage of keeping the ROV static is that the boundary conditions can be set up more easily for computer simulation. On the other hand, moving the ROV creates more meshes and thus is more complex to handle.

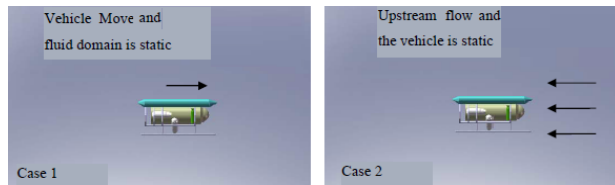


Fig.4 Physical model and computer simulation model use in CFD

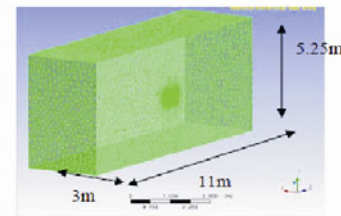
In most ROV operations, considering its scale and speed, the typical Reynolds number is greater than 1.0×10^6 , indicating turbulence in the flow field as shown in case 2 (see Fig.4). In turbulence flow, the fluid motion is characterized by a highly random, unsteady three-dimensional flow. The turbulent length and time scales are much smaller than the smallest finite volume meshes that are used in most numerical analyses. To generate such a small mesh is beyond the computing power currently available in the laboratory. The problem is solved by employing a turbulence model.

In general, turbulence models seek to modify the original unsteady Navier-Stokes equations with the Reynolds Averaged Navier-Stokes (RANS) equations. In the resulting Statistical Turbulence Model, each variable is separated into

a mean and a fluctuating component. The equations are time averaged over a time scale that is sufficiently longer than the time scale for the fluctuating components. The interval of the time scale is selected so that the resulting average term is independent of the length of the interval. ANSYS CFXTM was used because of the availability of the Shear-Stress-Transport (SST) model. Currently, the k- ω based Shear-Stress-Transport (SST) model is the most prominent two equations model to predict flow transition from laminar to turbulent. It was designed to give highly accurate predictions of the onset and the amount of flow separation under adverse eddy-viscosity. The process of using ANSYS-CFX in modeling the ROV's damping coefficients is shown below.

The boundary conditions and model environment setup must be established. The flow in the domain is expected to be turbulent and approximately isothermal. The SST turbulence model with automatic wall function treatment will be used because of its highly accurate predictions of flow separation. To take advantage of the SST model, the boundary layer should be resolved with at least 10 mesh nodes. This is done by inspecting the y^+ value on the surface of the ROV that must be around one.

The main difficulty is to determine the fluid domain dimension. In order to study damping force acting on the ROV in an unbound fluid domain, an infinite large fluid domain is needed. However, this is not practical both in CFD and even in the experiment. As shown in Fig.5, the fluid domain dimensions of around 20 times the length, width, and height of the ROV are used after the analysis is carried out to increase the domain size until the results start to vary. It is shown that the deviations in the results are negligible for these dimensions.



Setting	Value
Basic Settings > Fluids List	Water
Domain Models > Pressure > Reference Pressure	1 [atm]
Heat Transfer > Option	Isothermal
Heat Transfer > Fluid Temperature	20 [C]
Turbulence > Option	Shear Stress Transport

Fig. 5 Meshing for flow domain

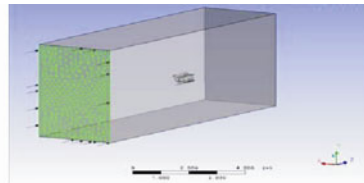
A flow speed is specified at the inlet boundary as shown in Fig.6. Physically, the flow upstream is uniform. It is set to be

normal to the plane and has a speed of 0.1m/s. Turbulence properties at the inlet flow are hard to specify because the turbulent kinetic energy and dissipation data are unknown. Therefore, their values have to be specified using sensible engineering assumptions and the effect of the choices made should be examined by sensitivity tests. In this study, the turbulent properties are specified using turbulent intensity.

To do this accurately it is good to have some form of measurements or previous experience. As shown below, common estimations of the incoming turbulence intensity are as follows.

- High-turbulence case: High-speed flows inside complex geometries such as heat exchangers and flow inside rotating machinery (turbines and compressors) typically have a turbulence intensity between 5% and 20%.
- Medium-turbulence case: Flows in not-so-complex devices such as large pipes and ventilation flows or low speed flows typically have a turbulence intensity between 1% and 5% .
- Low-turbulence case: External flow across cars, submarines, and aircraft in otherwise stationary fluid domains and flow generated by very high-quality wind tunnels typically have a turbulence intensity well below 1%.

The external flow over an underwater vehicle has a turbulence intensity well below 1%. Therefore, the turbulent intensity of 0.1% is selected for the ROV.

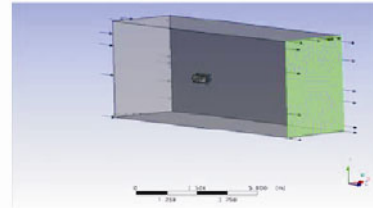


Setting	Value
Flow Regime > Option	Subsonic
Mass and Momentum > Option	Normal Speed
Mass and Momentum > Normal Speed	0.1 [m s ⁻¹]
Turbulence > Option	Intensity
Turbulence > Intensity	0.1%

Fig.6 Input settings

The outlet boundary condition is selected to have a weak influence on the upstream flow as shown in Fig.7. Specifying static pressure at the outlet plane is recommended (WS Atkins Consultants, 2002). At a high Reynolds Number, the wall-stress cannot be calculated correctly using the gradients on the coarse grids or meshes. The boundary layer is often too thin to be represented by the meshes. Hence, there is a

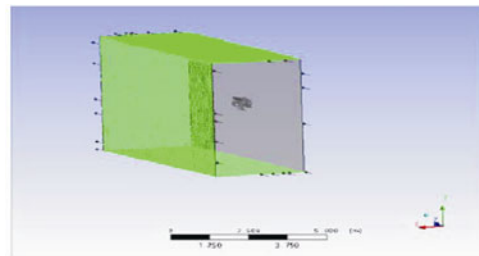
need to develop a model to approximate the wall-boundary condition used in ANSYS CFXTM (Launder *et al.*, 1974). In the log-law method, the near wall tangential velocity is logarithmically related to the wall shear stress. This relationship helps to predict the velocity profile in the boundary layer; hence, fine meshes are not necessary.



Setting	Value
Boundary Type	Outlet
Location	Outlet
Mass and Momentum > Option	Static Pressure
Mass and Momentum > Relative Pressure	0 [Pa]

Fig.7 Output settings

The top, bottom, and side surfaces of the rectangular fluid domain can be modeled using free slip wall boundary conditions as shown in Fig.8. On free slip walls, the shear stress is set to zero so that the fluid is not retarded. The velocity normal to the wall is also set to zero. However, it is not an ideal boundary condition, as the flow around the body can be affected by its close proximity to the walls.

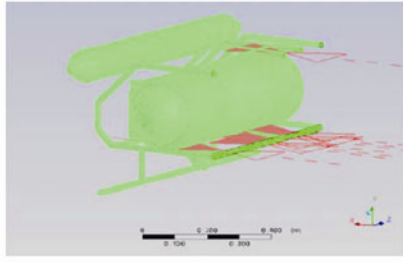


Setting	Value
Boundary Type	Wall
Location	FreeWalls
Wall Influence On Flow > Option	Free Slip

Fig.8 Free-slip boundary condition on fluid domain wall

Also, due to fluid viscosity, the flow speed on the vehicle's surfaces is zero. Hence, a non-slip wall boundary condition is applied on the vehicle's body as shown in Fig.9. As observed, the ROV has symmetry on the XZ plane. By using the

symmetry on the XZ plane, the computational time and resources are greatly reduced. On the symmetry plane, the gradients perpendicular to the plane are zero.



Setting	Value
Boundary Type	Wall
Location	Body
Wall Influence On Flow > Option	No Slip

Fig.9 Non-slip boundary condition on ROV surface

Initial values for all variables must be set before the computation begins. For a steady-state calculation, the initial variable values serve to give the ANSYS CFXTM Solver a flow field to start its calculations. Convergence is more rapidly achieved if sensible initial values are provided. However, converged results should not be affected by the initialization. Unless there is information about the turbulent eddy dissipation, it is recommended to use the automatic initial guess as shown in Table 2.

Table 2 Initial condition of the flow domain

Setting	Value
Initial conditions>	Automatic with
Cartesian velocity components>Option	value
Initial conditions>	0.1/(m·s ⁻¹)
Cartesian velocity components> U	
Initial conditions>	(Selected)
Turbulence eddy dissipation	

In the ANSYS-CFX's solver control, the selection of an appropriate time step size is essential to obtain good convergence rates. A physical time step is used to provide sufficient relaxation of the non-linearity so that a converged steady-state solution can be obtained. Physical time steps that are too large are characterized by oscillatory convergence or results that do not converge. Small time steps are characterized by very slow, steady convergence. For advection-dominated flows, the physical time step size is some fraction of a length scale divided by a velocity scale. A good approximation of the dynamical time for the flow is the time taken for the flow to make its way through the fluid domain. A reasonable estimate is around one third of the length of the fluid domain L and the mean velocity U . In this study, the time step is taken as:

$\Delta t = \frac{L}{3U} = \frac{10\text{m}}{3 \times 0.5\text{ms}^{-1}} = 6.67\text{s}$. For purpose of convenience, the time step is set to 5s as shown in Table 3.

Table 3 CFX's solver control

Setting	Value
Convergence control > Max iterations	60
Convergence control > Fluid timescale control > Timescale control	Physical timescale
Convergence control > Fluid timescale control > Physical timescale	5 s
Convergence criteria > Residual target	10 ⁻⁵

Before running the CFD simulation, face spacing that determines the surface area of a mesh in contact with the ROV's surface has to be defined. It must be small enough to capture the geometry of the ROV. The inflation layer is important in the capture of the boundary-layer flow and prediction of the flow separation. The number of inflation layer's increases on the surface of the ROV.

Fig.10 illustrates the effect of face spacing. One could observe that the mesh sizes increase exponentially away from the surface of the cylinder. It is important to have a smaller number of meshes far away from the body while keeping the meshes fine near to the body. The right-hand side of Fig.10 shows an enlarged image of the inflation layers. The boundary layer is contained within a thin layer near to the body. The inflation layer is thus important for capturing the flow near the boundary layer by providing a finer mesh in the region.

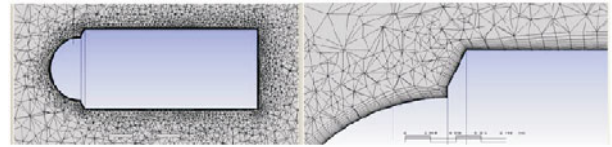


Fig.10 Cut surface mesh plot at the middle plane of the cylinder hull

The mesh domain around the ROV has around 2 219 437 elements. They consist of 1 067 375 tetrahedras, 1 140 046 wedges, and 12 016 pyramids as shown in Fig.11.

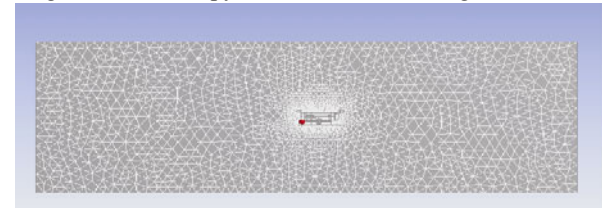


Fig.11 Meshing domain

The accuracy of the numerical approach was established through comparison of the current study with empirical and other computational results of the sphere. Flow at three different Reynolds Numbers has been selected for comparison. They are $Re = 100$, where the flow is in the

steady and axis-symmetric regime; $Re = 300$, where the flow is in the laminar periodic vortex-shedding regime; $Re = 1.1 \times 10^6$, where the flow is highly unsteady and the boundary layer may be turbulent. The mesh statistics and mesh settings used for the study are shown in Table 4.

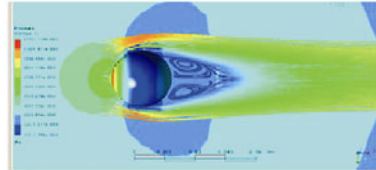
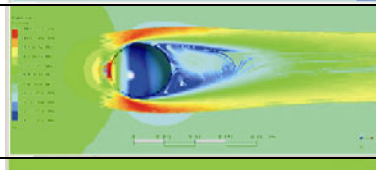
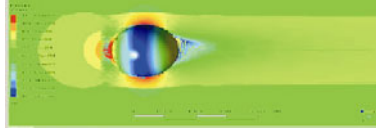
Table 5 presents the streamline plots of the sphere at different Reynolds numbers. The drag coefficients obtained in this study closely matched those in published results for Reynolds Number 100, 300, and 1.1×10^6 . This give the assurance that the procedures used in setting up the

simulation are done correctly. For the study of a sphere's drag, it is observed that the $y+$ value on the sphere surface must be close to unity.

Table 4 Mesh statistics and mesh settings

No of Mesh Elements	1255570
Face Spacing	0.001m ²
Face Spacing Expansion Factor	1.2
No of Inflation Layer	35
Thickness Multiplier	1
Inflation Expansion Factor	1.2

Table 5 Streamline plots for various Reynolds number (for sphere)

Reynolds Number	Source	Damping Coefficient (Cd)	Streamline Plots from CFD
100	Current study (using ANSYS-CFX TM)	1.26	
	(Kim and Choi, 2002)	1.09	
	Empirical	1.18	
300	Current study (using ANSYS-CFX TM)	0.814	
	(Johnson and Patel, 1999)	0.626	
	Empirical	0.826	
1.1×10^6	Current study (using ANSYS-CFX TM)	0.117	
	(Achenbach, 1972)	0.12-0.14	
	(Constantinescu et al., 2002)	0.084	

By applying the scale model, a similar phenomena as caused by the flow of the fluid around the body is formed. The region of disturbed flow (usually turbulent) at the downstream of the body is known as wake. As shown in the pressure distribution in Table 5, the wake creates a low pressure region at the rear end and thus high pressure in the front resists the vehicle motion. It exhibits similar phenomena as compared to the streamline plots for the sphere.

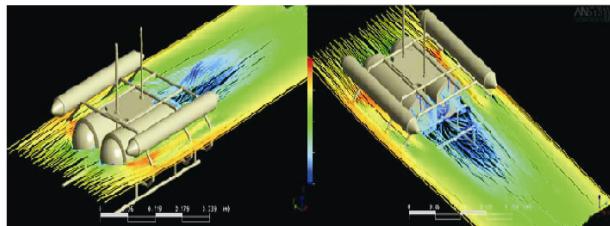


Fig.12 Streamline plot of the scaled model

Fig.13 shows the normal force and the tangential force acting on the scaled model. As observed, the forces converge to a constant value. It is an important indication that the flow has reached a steady state and thus is in accordance to the assumption of steady flow.

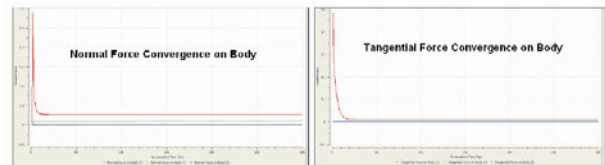


Fig.13 Iteration history of the normal force and tangential force acting on the scaled model

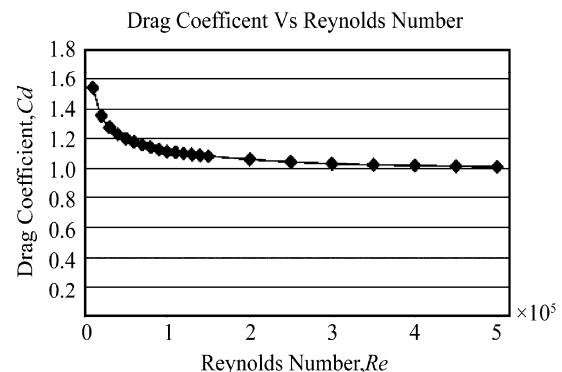


Fig.14 Drag coefficient versus Reynolds number

Furthermore, the computed drag coefficient converges to a

constant value as shown in Fig.14. The results imply that the drag coefficient is insensitive to the change of the Reynolds Number at the ROV operating range and can be treated as a constant drag coefficient. As observed in pressure distributions on the ROV body, the front of the ROV facing the flow experiences the largest pressure force. The total pressure difference between front and back causes the pressure drag, which is the dominant drag acting on a bluff body such as the ROV's body.

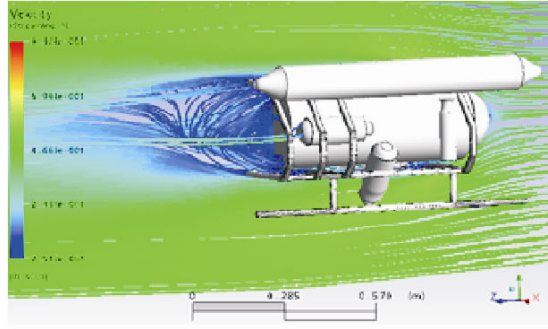
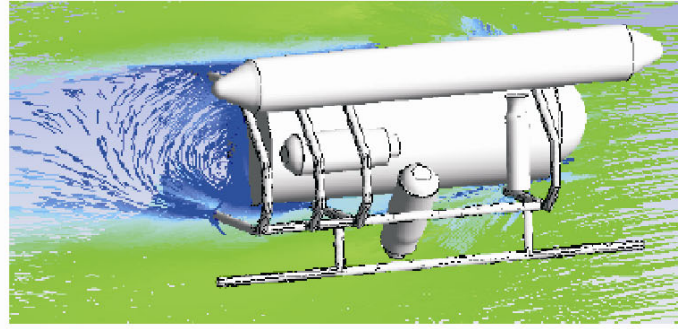


Fig.15 Streamline and vector plot at the center plane of the cylinder hull at flow speed: 0.5m/s



The hydrodynamic damping forces were computed by integrating the pressure on its surface. The drag forces in the surge, sway, and heave directions are plotted against the velocity in Figure 16. The drag value in heave is plotted between 0-0.2m/s because the ROV operates around this speed in actual operation. On the other hand, the vehicle operates between 0-0.5m/s in the surge and sway direction. As expected, the vehicle has the largest drag in the heave followed by the sway and surge. The heave has the largest frontal area normal to the flow direction of about 0.9m^2 compared to 0.5m^2 for sway and 0.45m^2 for surge. The drag in the sway is only slightly larger than the drag in the surge. This may be explained by the relative size of the frontal areas. From the second-order polynomial fit of the graph in Fig.16 and Fig.17, the linear damping and quadratic damping for respective directions are obtained. The results are subsequently tabulated in Table 7.

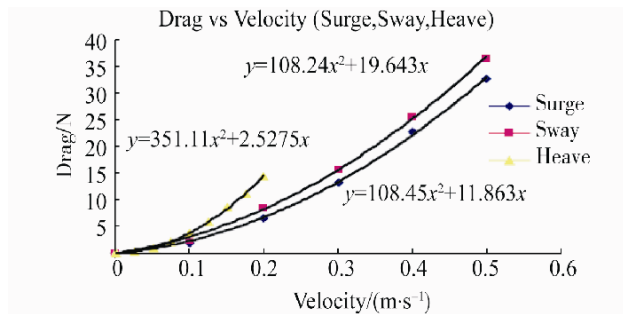


Fig.16 Drag force as the function of velocity in surge, sway and heave

Similar to the sphere case, the formation of wake at the rear of the ROV can be seen in Fig. 15. The wake or small regions representing the vortices of the flow separation increases in number and area. This turbulence region produces the nonlinear damping effect on the ROV. However, as the pitch angle is small, the nonlinear damping forces acting on the ROV can be quite small.

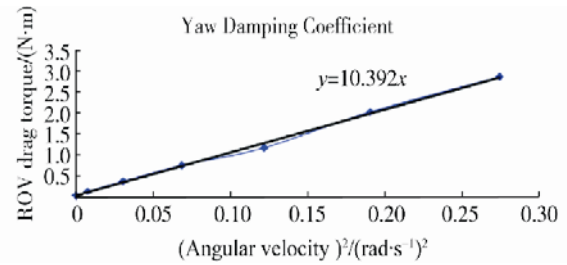


Fig.17 Drag torque as a function of square angular velocity

4 Experimental results from free-decaying method

In the previous sections, the parameters associated with the hydrodynamic forces acting on the ROV were estimated using CFD simulations such as ANSYS-CFX™. However, as stated in the CFD guidelines (WS Atkins Consultants, 2002) these parameter values should be corroborated by other means. The accuracy of the results can only be truly determined if the results can be independently confirmed using a different approach. Therefore, in this section, verification of the drag force coefficients are performed by comparing the value predicted by CFD and that to be obtained experimentally using the scaled down model of the ROV (which hereafter will be referred to as the scaled model). By applying laws of Similitude (Conte *et al.*, 2004), the hydrodynamics parameters of the scaled model can be scaled up to predict the corresponding values for the true model (actual ROV model). The method is a viable alternative for estimating some pertinent hydrodynamic parameters without extensive and expensive facilities and instrumentation at a very early stage of the ROV development.

The hydrodynamic parameters can be extracted from an experiment as shown in Fig.18, in which the scale model attached at the end of the rod performs a free decay motion in water. The details of the experiments and results can be found in (Eng, 2007; Eng *et al.*, 2009; Eng *et al.*, 2008). The least-square approach was used to determine the hydrodynamic damping parameters.

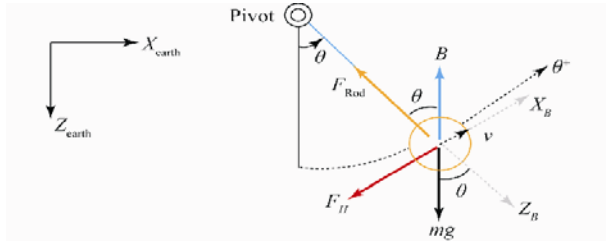


Fig.18 Free body diagram of the setup

To perform the experiments according to the concept shown in Fig.18, experiments were carried out in a 0.3m×0.2m×0.3m open tank (see Fig.19) giving a characteristics length ratio of about 0.3:1 for the ROV to investigate the motion characteristics of the scaled ROV in surge, sway, heave, and yaw in the positive direction. The scaled ROV is attached to one end of a pendulum in the water and fully submerged in the water by an aluminum frame mounted on the water tank. The pendulum is set in motion in an arc at a certain radius. With a small black mark on the rod, the motion of the pendulum can be captured by a video camera. The recorded trajectory of the black mark can be digitized using the open-source program VirtualDubMod (Ng, 1998). For each frame, the *X* and *Y* coordinates of the black mark were acquired for image processing using the MATLABTM Image Processing Toolbox. The camera is used to record the pendulum motion and the video is split into multiple frames with up to 30 frames per second. Fig.20 and Fig.21 show some image frames of the free decay motion of the pendulum in the surge and yaw directions. In the earth-fixed frame, the scaled model is constrained to rotate in one plane about the pivot point.



Fig.19 RRC ROV prototype in water tank (orientated in surge direction)

A similar test in the heave direction, with the scaled model rotated 90° facing the direction of the motion, was performed to estimate the parameters in the yaw direction. In order to identify the parameters of the yaw motion, the pendulum's rod was replaced by a torsion spring. The image sequence of the scaled model in the yaw direction can be seen in Fig.21. The scaled model exhibits pure rotational motion in the water.

Each experiment was repeated a number of times at each velocity and the average damping force readings (the longitudinal force, the transverse force, altitudinal force, and moment are the surge force, sway force, heave force, and yaw moment, respectively) were obtained. As a result, the curves of the thrust versus the motion variables *u*, *v*, *w*, and *r* were obtained. With the vehicle surge, sway, or heave set to a constant velocity, the thrust can be considered to be equal to the hydrodynamic force, and the moments can be considered to be equal to the hydrodynamic moments when it yaws at a constant angular velocity. Therefore, the curves of the thrust versus the motion variables are assumed to be same as the curves of hydrodynamic loads versus these motion variables. As the hydrodynamics forces resist the motion, the amplitude of the swing will decay slowly over time. The hydrodynamics damping coefficients can then be determined using the least-square method in offline situations.

In the experiment, the amplitude of the wave created by the ROV is smaller than the amplitude created by the pendulum itself. As seen in Table 6a to 6d, the changes in the velocity (i.e. the speed of the free-decaying test) do not significantly affect the hydrodynamic parameters. In this respect, the test regime is still within the “steady-state” with small waves contributed by the scaled ROV at a low speed of 0.55m/s (maximum). Based on the root mean squared errors (RMS) computations, the values that correspond to the lowest and consistent errors are chosen. As a result, the hydrodynamic damping parameters at velocity 0.55 m/s (for surge), 0.50 m/s (for sway), 0.35 m/s (for heave), and 0.55 rad/s (for yaw) are used.

To obtain the actual results from the scale model, the following scaling method is used for scaled-up results obtained from the experiments results as tabulated in Table 6a to 6d. For the drag forces, this can be done using the laws of Similitude. The magnitude of the drag force acting on a submerged object moving in a rectilinear direction is given by:

$$D = \frac{1}{2} \rho C_d A U^2 \quad (9)$$

where ρ is the fluid density, A is the frontal area, U is the forward velocity, and C_d is the drag coefficient, which is a function of Reynolds Number, Re (Hoerner, 1965) for a submerged vehicle.

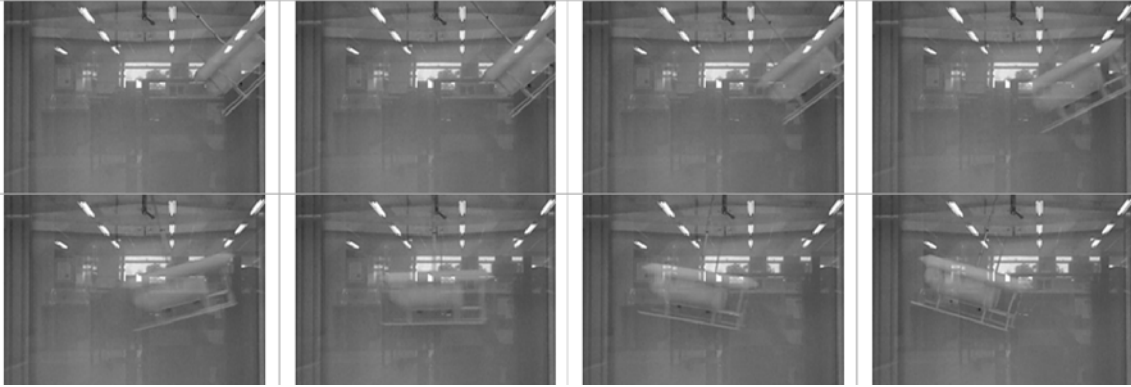


Fig.20 Image Sequence of the scaled model under pendulum motion—in surge direction

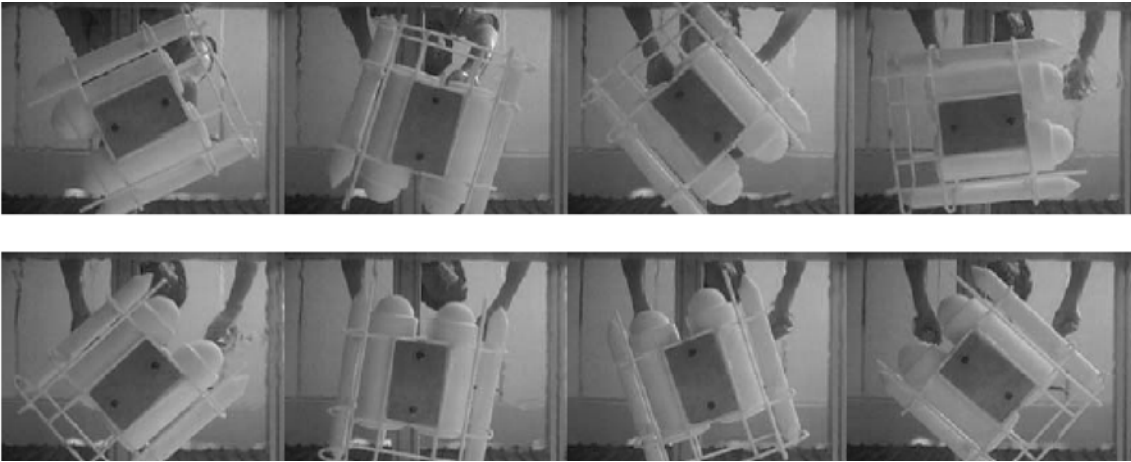


Fig.21 Image sequence of the scaled model in the yaw direction

Table 6a Hydrodynamic damping parameters at different velocity(surge direction)

Direction	Max. Speed $/(m \cdot s^{-1})$	K_L	K_Q	RMS error
Surge	0.55	1.2736	9.9392	0.0568
		1.7768	9.1907	0.0500
		1.6040	9.5018	
		(Avg: 1.550)	(Avg: 9.540)	0.0552
	0.50	1.2011	9.667	0.0572
		1.5800	9.008	0.0685
		1.4123	8.995	0.0717
	0.35	1.0192	7.0113	0.0632
		0.9988	7.0393	0.0541
		1.0078	7.4931	0.0797

Table 6b Hydrodynamic damping parameters at different velocity(sway direction)

Direction	Max. Speed $/(m \cdot s^{-1})$	K_L	K_Q	RMS error
Sway	0.55	3.3456	8.0991	0.0266
		3.4577	7.9981	0.0372
		3.5678	8.0113	0.0213
		3.1388	7.8691	0.0277
	0.50	3.6694	7.4974	0.0234
		3.4671	7.3418	0.0269
		(Avg: 3.420)	(Avg: 7.560)	
	0.35	3.0069	6.1170	0.0294
		2.9977	7.0008	0.0333
		2.8997	7.0912	0.0459

Table 6c Hydrodynamic damping parameters at different velocity (heave direction)

Direction	Max. Speed /(m·s ⁻¹)	K_L	K_Q	RMS error
Heave	0.55	8.0711	10.0125	0.0240
		8.1113	10.0034	0.0331
		7.9871	9.9874	0.0423
	0.50	8.0212	9.0982	0.0341
		7.8971	9.0124	0.0556
		7.8132	9.1492	0.0421
	0.35	6.5011	8.9937	0.0239
		6.8327	8.1637	0.0219
		6.1905	11.035	0.0277
		(Avg: 6.520)	(Avg: 9.390)	

Table 6d Hydrodynamic damping parameters at different velocity (for yaw direction)

Direction	Max. Speed /(rad·s ⁻¹)	K_L	K_Q	RMS error
Yaw	0.35	0.00036	0.012	0.0311
		0.0012	0.009	0.0425
		0.0027	0.0074	0.0511
	0.50	0.007	0.0011	0.0427
		0.0024	0.0092	0.0588
		0.0017	0.0084	0.0403
	0.55	0.00073	0.0220	0.0320
		0.0031	0.0180	0.0300
		0.0054	0.0150	0.0380
		(Avg: 0.003)	(Avg: 0.019)	

Table 7 Damping coefficients obtained from free decay experiment and ANSYS-CFXTM

Methods	Damping Coefficients							
	Surge (0–0.5m/s)		Sway (0–0.5m/s)		Heave (0–0.2 m/s)		Yaw (0–0.5 rad/s)	
	K_L	K_Q	K_L	K_Q	K_L	K_Q	K_L	K_Q
ANSYS-CFX TM	11.863	108.45	19.640	108.24	2.3756	351.98	0	10.390
Experiment (Scaled-up)	17.240	106.03	38.060	84.100	72.530	104.41	1.180	7.5100
Experiment (Scaled)	1.55	9.54	3.42	7.56	6.52	9.39	0.003	0.019

The scaled model was built with a characteristic length of about 1/0.3 of the ROV to give geometric similarity. It was tested at a speed of about 0.30–0.5m/s to give the same Re of about 1.6×10^5 as the ROV operated at a speed of about 0.3m/s in sea-water. This maintains the dynamic similitude between the models. As shown in Fig.14, the drag coefficient is nearly constant for a range of Reynolds Numbers from 2.0×10^5 to about 5.0×10^5 . Using Eq.(9), the ratio of drag force experienced by the real or actual model of the ROV (subscript a) to the damping force experienced by the scaled model (subscript s) can be obtained.

$$\frac{D_a}{D_s} = \frac{\rho_a C_d A_a U_a^2}{\rho_s C_d A_s U_s^2} \quad (10)$$

The drag force in the rotational direction (Presterio, 2001) can be scaled up as shown.

$$T_d = \sum_i r_i f_{di} = \frac{1}{2} C_d \rho A_p L^3 \dot{\theta}^2$$

$$\frac{T_a}{T_s} = \frac{\rho_a C_d A_a L_a^3 \dot{\theta}_a^2}{\rho_s C_d A_s L_s^3 \dot{\theta}_s^2} \quad (11)$$

As a sanity check on the method, the ANSYS-CFXTM study of the scaled model for similar flow condition was included. In the CFD test using ANSYS-CFXTM, the flow in the domain surrounding the vehicle was simulated for various speeds as shown in Table 7, and the hydrodynamic loads were obtained by integrating the pressure on its surface, which was obtained via the simulation.

The experimentally or theoretically identified quadratic damping parameters are tabulated in Table 7. The average readings at the respective velocities are used. Both the theoretical and experimental results indicate that the damping is the smallest in the surge DOF and largest in the heave DOF. This is reasonable as the vehicle's cross-section area is smallest in the surge DOF and largest in heave DOF. However, the vehicle's linear damping behaves in the opposite manner with a smaller value. This is due to the skin friction being more dominant in the surge DOF than in the heave DOF. Moreover, it is difficult to replicate the same surface roughness of the actual ROV model on the scaled model. This can be seen in that the values of the linear

damping coefficients are generally less than the quadratic damping coefficients. However, the quadratic damping forces are lower than the linear damping forces due to the “square” of the velocity terms.

As shown in Table 7, the forces show good agreement when the vehicle surges and sways; however, the calculated heave and moments are greater than the experimental results. There are two main reasons for the differences. The first is that the thruster is not considered in the CFD model, which may lead to an error in the simulated hydrodynamic center of rotation and heave motion. The hydrodynamic interaction between the thrusters and the rest of the body is strong, so the effects of the thrusters while the thrusters are operating must be considered. Detailed work on the effects of the operating thrusters in CFD will need to be carried out in future work. The other is that some small changes were made to adjust the centre of gravity and buoyancy in the experiments due to the fixture that held the ROV during the test, which implies that they may be different from their design points. These factors have some effect on the dynamic of the vehicle, whereas their effect on the translational dynamics is much smaller than the rotational and vertical dynamics. In addition, it must be noted that this vehicle was operated in the pendulum rig set up with some pitching effect during the experiment, and therefore, the pitching may also increase the experimental values. This can be seen in the slight increase over the experiment results.

The graphical plots of the drag force can be seen in Fig.22 to Fig.25. Comparison should not be done on the basic K_L versus K_Q . Indeed, only the combination effect of both should be compared by plotting the total drag force within the velocity range of operation. Fig.22 and Fig.23 show that the drag forces for surge and sway match closely with values from CFX, although they are consistently larger. The main reason is that the smooth surface is assumed in ANSYS-CFXTM computation, and this probably underestimates the drag value due to skin friction. For the heave direction, the experimental values are consistently larger than the ANSYS-CFXTM values by a constant for velocity range from 0.1 m/s to 0.2m/s. This is expected because there are two rectangular metal plates attached to the scaled model which increases the drag in the heave direction (as can be seen from Fig.24). In the ANSYS CFXTM study, the two iron plates do not exist in the full-scale ROV.

The results for the yaw direction from both the experiment and the ANSYS CFXTM are closely matched as shown in Fig.25. This is an unexpected result as calculating the drag moment is considered difficult by the CFD community. The experiment data of the ROV was obtained by scaling up the drag moment from scaled model tests that involve a very big scaling factor. Therefore, any error in scaled model tests will inevitably be scaled up. This is probably a limitation in the use of the free decay test for a scale model. Nevertheless, the magnitude of the moment drag is reasonable and could be a

good estimation of the true moment drag that is experienced by the ROV.

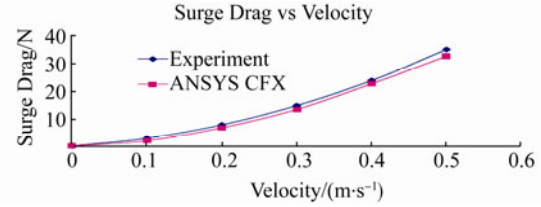


Fig.22 Comparison of damping force along surge direction

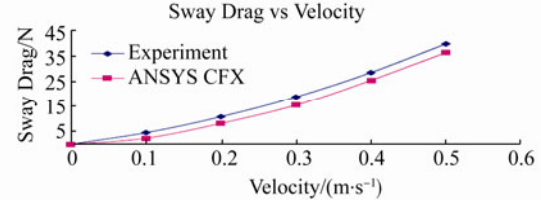


Fig.23 Comparison of damping force along sway direction

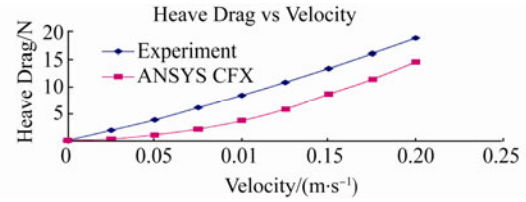


Fig.24 Comparison of damping force along heave direction

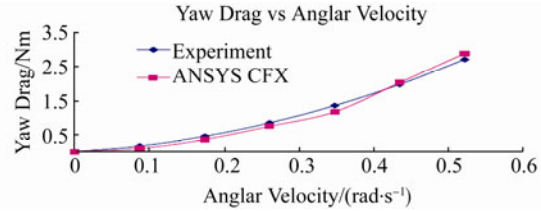


Fig.25 Comparison of damping force in yaw direction

In summary, the measured values of the pendulum trajectory in water and the scale model using the estimated parameters and the theoretical results compare quite well with the experimentally measured responses both in magnitude and trend. Hence, the linear hydrodynamic damping matrix is considered in the subsequent control and can be written as:

$$\mathbf{D} = -\text{diag}\{17.20, 38.06, 72.50, 1.665, 1.456, 1.180\}$$

5 Conclusions

In this paper, the modeling of the hydrodynamic damping of a complex-shaped remotely-operated vehicle (ROV) for pipeline tracking was shown. The dynamic modeling involved the proposed semi-predictive method of using the computational fluid dynamic software ANSYS-CFXTM on the ROV model obtained from PRO/ENGINEERTM and the experimental tests using the free-decaying method. The method was shown to be a viable alternative for estimating pertinent hydrodynamic parameters without extensive and expensive facilities and instrumentation at a very early stage of the development of complex-shaped underwater robotic

vehicles. Experiment results were observed to coincide with the simulated results from ANSYS-CFXTM. Hence, the proposed method provides a convenient platform of graphical modeling using PRO/ENGINEERTM and the computational fluid dynamic ANSYS-CFXTM to compute the damping coefficients of complex-shaped underwater robotic vehicles such as ROVs.

References

- Jalving B (1994). The NDRE-AUV flight controls system. *IEEE Journal of Oceanic Engineering*, **19**(4), 497-501.
- Healey AJ, Lienard D (1993). Multivariable sliding mode control for autonomous diving and steering of unmanned underwater vehicles. *IEEE Journal of Oceanic Engineering*, **18**(3), 327-339.
- Gomes RMF, Sousa JB, Pereira FL (2003). Modelling and control of the IES project ROV. *Proceedings of European Control Conference*, Cambridge, UK, 1-6.
- Antonelli G, Chiaverini S, Sarkar N, West M (2001). Adaptive control of an autonomous underwater vehicle: Experimental results on ODIN. *Transactions on Control Systems Technology*, *IEEE*.
- Fjellstad OE, Fossen TI (1994). Position and attitude tracking of AUVs: A quaternion feedback approach. *IEEE Journal of Oceanic Engineering*, **19**(4), 512-518.
- Fossen TI (1994). *Guidance and control of ocean vehicles*. John Wiley & Sons Ltd.
- Fossen TI (2002). Marine control systems; guidance, navigation and control of ships. Rigs and underwater vehicles. *Marine Cybernetics*.
- Goheen KR (1991). Modeling methods for underwater robotic vehicle dynamics. *Journal of Robotic Systems*, **8**(3), 295-317.
- An PE, Folleco A (2003). Modeling and simulation of autonomous underwater vehicles: design and implementation. *IEEE Journal of Oceanic Engineering*, **28** (2), 283-296.
- Gomes RMF, Sousa JB, Pereira FL (2003). Modelling and control of the IES project ROV. *Proceedings of European Control Conference*, Cambridge, UK.
- Goodman A (1960). Experimental techniques and methods of analysis used in submerged body research. *Third Symposium on Naval Hydrodynamics*, Office of Naval Research.
- Williams CD, Mackay M, Perron C, Muselet C (2000) The NRC-IMD marine dynamic test facility: A six-degree-of-freedom forced-motion test apparatus for underwater vehicle testing. *International UUV Symposium*, Newport, RI, 1-6.
- Jones DA, Clarke DB, Brayshaw IB (2002). The calculation of hydrodynamic coefficients for underwater vehicles. DSTO Platforms Sciences Laboratory, Fishermans Bend, Australia, Report. DSTO-TR-1329.
- Sarkar T, Sayer PG, Fraser SM (1997). A study of autonomous underwater vehicle hull forms using computational fluid dynamics. *International Journal for Numerical Methods in Fluids*, **25**(11), 1301-1313.
- Tyagi A, Sen D (2006). Calculation of transverse hydrodynamic coefficients using computational fluid dynamic approach. *Ocean Engineering*, **33**(5), 798-809.
- Wilson R, Paterson E, Stern F (2006). Unsteady RANS CFD method for naval combatant in waves. *Proceedings of the 22nd ONR Symposium on Naval Hydrodynamics*, Washington DC, 532-549.
- WS Atkins Consultants (2002). Best Practices Guidelines for Marine Applications of CFD, MARNET-CFD Report.
- Conte G, Zanolini SM, Scaradozzi D, Conti A (2004). Evaluation of hydrodynamics parameters of a UUV. A preliminary study, International Symposium on Control. *Communications and Signal Processing, ISCCSP*, Hammamet.
- MSS. Marine Systems Simulator (2010). Viewed 26.06.2011, <http://www.marinecontrol.org>.
- Eng YH (2007) Identification of hydrodynamic terms for underwater robotic vehicle. Master First Year Report, NTU, Robotic Research Center, Mechanical and Aerospace Engineering.
- Eng YH, Lau WS, Low E, Seet GL, Chin CS (2009). A novel method to determine the hydrodynamic coefficients of an eyeball ROV. *AIP Conference Proceedings*, 1089, 11-22.
- Eng YH, Lau WS, Low E, Seet GL, Chin CS (2008). Estimation of the hydrodynamic coefficients of an ROV using free Decay Pendulum motion. *Engineering Letters*, **16**(3), 326-331.
- Lauder BE, Spalding DB (1974) The numerical computation of turbulent flows. *Comp. Methods Appl. Mech. Eng.*, **3**, 269-289.
- Kim D, Choi H (2002). Laminar flow past a sphere rotating in the stream wise direction. *Journal of Fluid Mechanics*, **461**, 365-386.
- Johnson TA, Patel VC (1999). Flow past a sphere up to a Reynolds number of 300. *Journal of Fluid Mechanics*, **378** (1), 19-70.
- Achenbach E (1972). Experiments on the flow past spheres at very high Reynolds number. *Journal of Fluid Mechanics*, **54**, 565-575.
- Constantinescu GS, Pacheco R, Squires, KD (2002). Detached-Eddy simulation of flow over a sphere. *AIAA, Aerospace Sciences Meeting*, Paper 2002-0425.
- Hoerner SF (1965). Fluid-dynamic drag: Practical information on aerodynamic drag and hydrodynamic resistance. *Hoerner Fluid Dynamics*, Washington.
- Presterio T (2001). Verification of a six-degree of freedom simulation model for the REMUS autonomous underwater vehicle. Master's thesis, Mechanical and Oceanographic Engineering, Massachusetts Institute of Technology and the Woods Hole Oceanographic Institution.
- Ng EYK, Tan CK (1998). Viscous flow simulation around a moving projectile and URV. *International Journal of Computer Applications in Technology*, **11**, 350-362.



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